Linux Lab Report

Directories, regular files, links, users, and groups management

April 6, 2026

Part 1 – Directory Structure

Commands Executed

Create hour main workspace directory called « techcoorp » under the root directory

```
mia@fedora:~$ mkdir techcoorp
mia@fedora:~$
```

Create the following subdirectories « dev », « test », « manager », and « shared » under the « techcoorp » directory.

```
mia@fedora:~$ mkdir techcoorp/dev techcoorp/test techcoorp/manager techcoorp/shared
mia@fedora:~$
```

Verify structure

```
mia@fedora:~$ ls -l techcoorp/
total 0
drwxr-xr-x. 1 mia mia 0 Apr  6 09:45 dev
drwxr-xr-x. 1 mia mia 0 Apr  6 09:45 manager
drwxr-xr-x. 1 mia mia 0 Apr  6 09:45 shared
drwxr-xr-x. 1 mia mia 0 Apr  6 09:45 test
mia@fedora:~$
```

Part 2 – File Manipulation

Commands Executed

Create the following files: «dev/code.txt», «test/test_cases.txt», and «manager/plan.txt »

```
mia@fedora:~$ touch techcoorp/dev/code.txt techcoorp/test/test_cases.txt techcoorp/manager/plan.txt
mia@fedora:~$
```

Add content in all the created files:

```
mia@fedora:~$ echo "dev code" > techcoorp/dev/code.txt
mia@fedora:~$ echo "test cases" > techcoorp/test/test_cases.txt
mia@fedora:~$ echo "project plan" > techcoorp/manager/plan.txt
mia@fedora:~$
```

Display the content of each file:

```
mia@fedora:~$ cat techcoorp/dev/code.txt
dev code
mia@fedora:~$ cat techcoorp/test/test_cases.txt
test cases
mia@fedora:~$ cat techcoorp/manager/plan.txt
project plan
mia@fedora:~$
```

Copy the « plan.txt » file into the « shared » directory and move the «test_cases.txt» file under the « shared » directory.

```
mia@fedora:~$ cp techcoorp/manager/plan.txt techcoorp/shared/
mia@fedora:~$ mv techcoorp/test/test_cases.txt techcoorp/shared/
mia@fedora:~$
```

Rename the « code.txt » file to « app_code.txt »

```
mia@fedora:~$ mv techcoorp/dev/code.txt techcoorp/dev/app_code.txt
mia@fedora:~$ ls -l techcoorp/dev/
total 4
-rw-r--r--. 1 mia mia 9 Apr  6 09:51 app_code.txt
mia@fedora:~$
```

Part 3 – Users & Groups

Commands Executed

Create the following groups: (i) developers (ii) testers (iii) managers

```
mia@fedora:~$ sudo groupadd developers
[sudo] password for mia:
mia@fedora:~$ sudo groupadd testers
mia@fedora:~$ sudo groupadd managers
mia@fedora:~$
```

Create the following users: (i) dev1 (ii) dev2 (iii) test1 (iv) manager1

```
mia@fedora:~$ sudo useradd dev1
mia@fedora:~$ sudo useradd dev2
mia@fedora:~$ sudo useradd test1
mia@fedora:~$ sudo useradd manager1
mia@fedora:~$
```

Assign users to groups:

```
mia@fedora:~$ sudo usermod -aG developers dev1
[sudo] password for mia:
mia@fedora:~$ sudo usermod -aG developers dev2
mia@fedora:~$ sudo usermod -aG testers test1
mia@fedora:~$ sudo usermod -aG managers manager1
mia@fedora:~$
```

Set passwords for all the users:

```
mia@fedora:~$ sudo passwd dev1
[sudo] password for mia:
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
mia@fedora:~$ sudo passwd dev2
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
mia@fedora:~$ sudo passwd test1
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
mia@fedora:~$ sudo passwd manager1
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: password updated successfully
mia@fedora:~$
```

Verify that all users and groups are correctly created.

```
mia@fedora:~$ cat /etc/passwd
dev1:x:1004:1007::/home/dev1:/bin/bash
dev2:x:1005:1008::/home/dev2:/bin/bash
test1:x:1006:1009::/home/test1:/bin/bash
manager1:x:1007:1010::/home/manager1:/bin/bash
mia@fedora:~$
```

```
mia@fedora:~$ cat /etc/group
developers:x:1004:dev1,dev2
testers:x:1005:test1
managers:x:1006:manager1
dev1:x:1007:
dev2:x:1008:
test1:x:1009:
manager1:x:1010:
mia@fedora:~$
```

Part 4 – Links

Commands Executed

Create a hard link called « plan_hard.txt » that refers to « plan.txt ».

```
mia@fedora:~$ ln techcoorp/manager/plan.txt techcoorp/manager/plan_hard.txt
mia@fedora:~$
```

Create a symbolic link called « test_link.txt» that refers to « test_cases.txt».

```
mia@fedora:~$ ln -s techcoorp/shared/test_cases.txt techcoorp/test/test_link.txt
mia@fedora:~$
```

Verify that the links are created.

```
mia@fedora:~$ ls -li techcoorp/manager/plan.txt techcoorp/manager/plan_hard.txt
19011 -rw-r--r--. 2 mia mia 13 Apr  6 09:52 techcoorp/manager/plan_hard.txt
19011 -rw-r--r--. 2 mia mia 13 Apr  6 09:52 techcoorp/manager/plan.txt
mia@fedora:~$ ls -li techcoorp/test/test_link.txt
19418 lrwxrwxrwx. 1 mia mia 31 Apr  6 10:39 techcoorp/test/test_link.txt -> techcoorp/shared/test_cases.txt
mia@fedora:~$
```

Experiment:

```
mia@fedora:~$ rm techcoorp/manager/plan.txt
mia@fedora:~$ cat techcoorp/manager/plan_hard.txt
project plan
mia@fedora:~$ ls -li techcoorp/manager/plan_hard.txt
19011 -rw-r--r--. 1 mia mia 13 Apr  6 09:52 techcoorp/manager/plan_hard.txt
mia@fedora:~$ rm techcoorp/shared/test_cases.txt
mia@fedora:~$ cat techcoorp/test/test_link.txt
cat: techcoorp/test/test_link.txt: No such file or directory
mia@fedora:~$ ls -li techcoorp/test/test_link.txt
19418 lrwxrwxrwx. 1 mia mia 31 Apr  6 10:39 techcoorp/test/test_link.txt -> techcoorp/shared/test_cases.txt
mia@fedora:~$
```

Part 5 – Mini Challenge

Commands Executed

Create the new user « intern1 » with a specific home directory called « intern_home »

```
mia@fedora:~$ sudo useradd -d /home/intern_home intern1
[sudo] password for mia:
mia@fedora:~$
```

Add the « intern1 » user to « developers » group.

```
mia@fedora:~$ sudo usermod -aG developers intern1
mia@fedora:~$ id intern1
uid=1008(intern1) gid=1011(intern1) groups=1011(intern1),1004(developers)
mia@fedora:~$
```

Display all users on the system: Which file contains this information?

```
mia@fedora:~$ cat /etc/passwd
```

Display all groups: Which file contains this information?

```
mia@fedora:~$ cat /etc/group
```

Change the username of the « intern1 » to « intern2 »

```
mia@fedora:~$ sudo usermod -l intern2 intern1
mia@fedora:~$
```

Add« intern2 » to another group (« testers ») without removing existing groups

```
mia@fedora:~$ sudo usermod -aG testers intern2
mia@fedora:~$ id intern2
uid=1008(intern2) gid=1011(intern1) groups=1011(intern1),1004(developers),1005(testers)
mia@fedora:~$
```

Lock the « intern2 » account.

```
mia@fedora:~$ sudo passwd intern2
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
mia@fedora:~$ sudo usermod -L intern2
mia@fedora:~$
```

Unlock the « intern2 » account.

```
mia@fedora:~$ sudo usermod -U intern2
mia@fedora:~$
```